

AI driven Market sentiment analysis & News summarisation

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Agenda

- **Executive Summary**
- **Business Problem**
- **Exploratory Data Analysis**
- **Part 1: Sentiment Analysis**
 - Text vectorisation
 - Model Building
 - Model Performance Summary
- **Part 2: News summarisation**
 - Data pre-processing
 - Prompt engineering
 - News summarisation
- **Lessons & Conclusion**

Executive summary

- **SBERT is Promising:** Among the base models, Random Forest with SBERT embeddings showed the best validation performance, suggesting that the rich contextual embeddings from Sentence Transformers are more effective for this task than traditional Word2Vec or GloVe.
- **Tuning Needs Improvement:** The current tuning efforts for the Random Forest with SBERT embeddings have not yet yielded an improvement in generalization. The F1-score decreased from ~0.505 (base) to ~0.418 (tuned). This suggests that the hyperparameter search space might still be too restrictive, or the found parameters are not truly optimal for generalization.
- **Small Validation Set:** The small size of the validation set (21 samples) makes the validation metrics highly sensitive and potentially unstable, making it challenging to reliably assess generalization performance during tuning.
- **News Extraction and Categorization:** The model successfully processed weekly news articles and categorized key events into 'Positive Events' and 'Negative Events' as structured lists. This demonstrates the effectiveness of the LLM in understanding and segmenting news sentiment.

Business Problem

- Investment firms and financial analysts struggle to stay updated with the vast volume of news and opinions from a wide variety of sources
- Market sentiment plays a significant role in the price of stocks of companies listed under the stock exchange.
- Investment firms need sophisticated tools to analyze market sentiment, its impact on the market and integrate this information into their investment strategies.

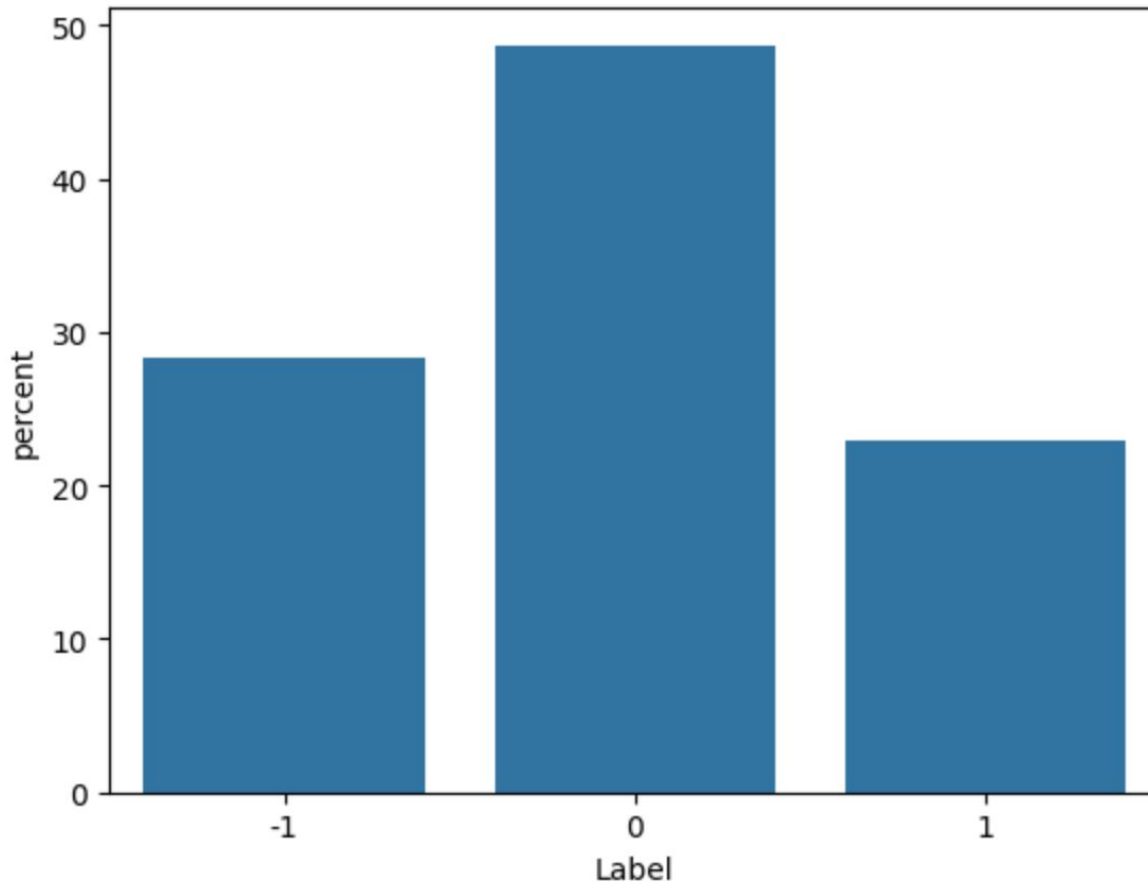
Project goal

- To help investment firms and financial analysis to interpret stock related news and their impact on their investment strategies
- Develop an AI-driven sentiment analysis system that will automatically process and analyze news articles to gauge market sentiment, and summarize the news at a weekly level.
- This will empower their financial analysts with actionable insights, leading to more informed investment decisions and improved client outcomes.

Dataset

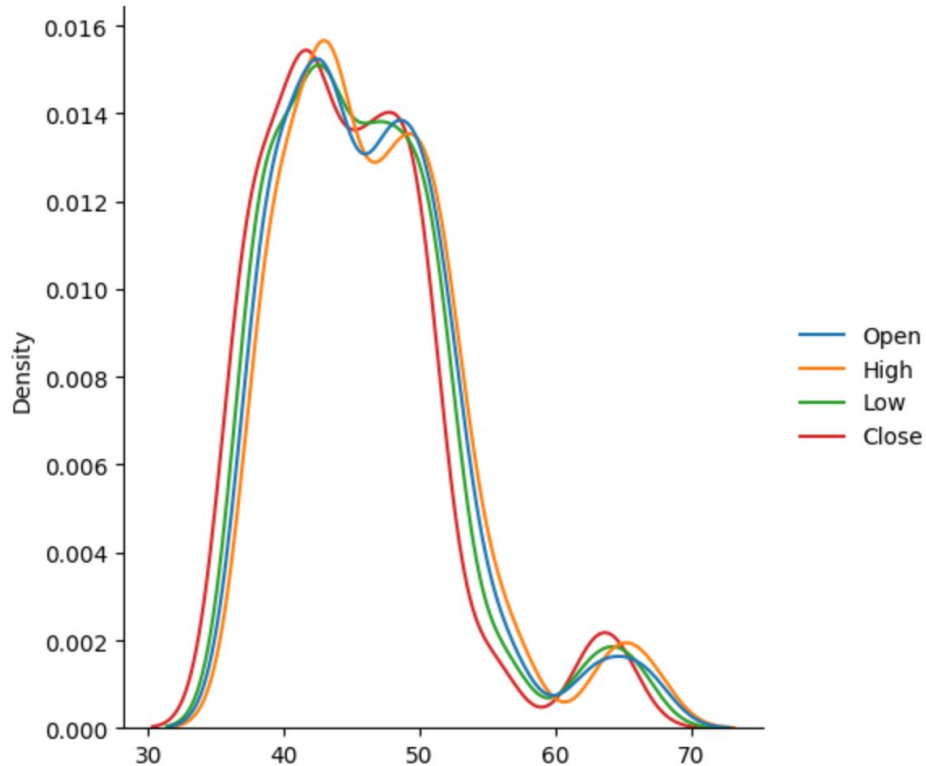
- 349 news articles for a specific company listed under NASDAQ, along with data on its daily stock price and trade volumes.
 - **Date** : The date the news was released
 - **News** : The content of news articles that could potentially affect the company's stock price
 - **Open** : The stock price (in \$) at the beginning of the day
 - **High** : The highest stock price (in \$) reached during the day
 - **Low** : The lowest stock price (in \$) reached during the day
 - **Close** : The adjusted stock price (in \$) at the end of the day
 - **Volume** : The number of shares traded during the day
 - **Label** : The sentiment polarity of the news content
 - 1: positive
 - 0: neutral
 - 1: negative

EDA: Class distribution



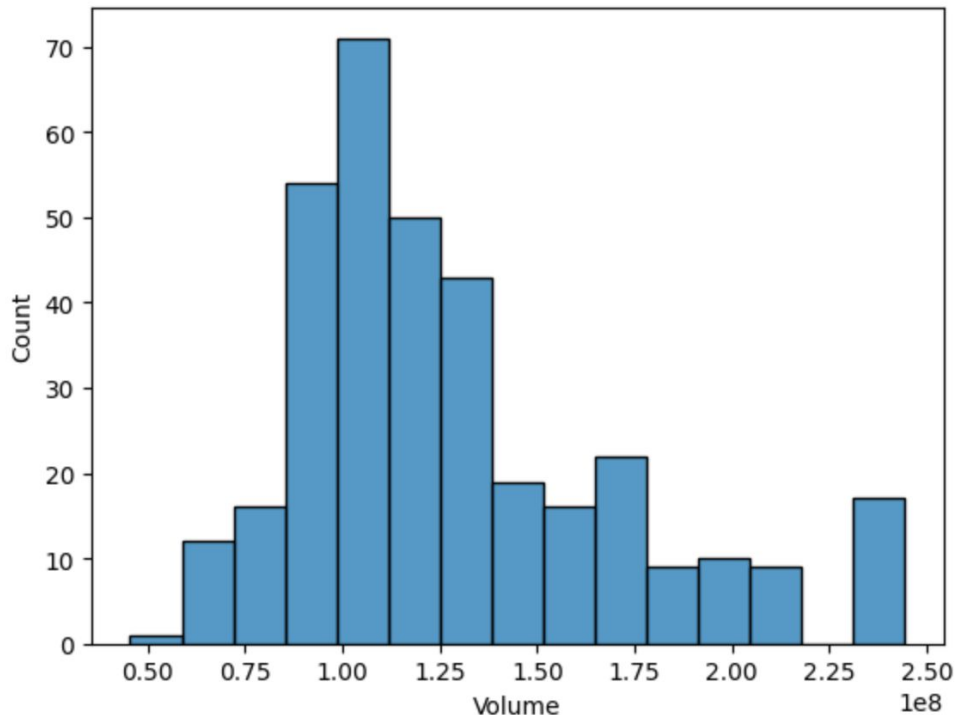
- 48% of the news articles are neutral (0), 24% are negative (-1), and roughly 28% are positive (1).
- Neutral sentiment is the most frequent category.

EDA: Density plot of Price



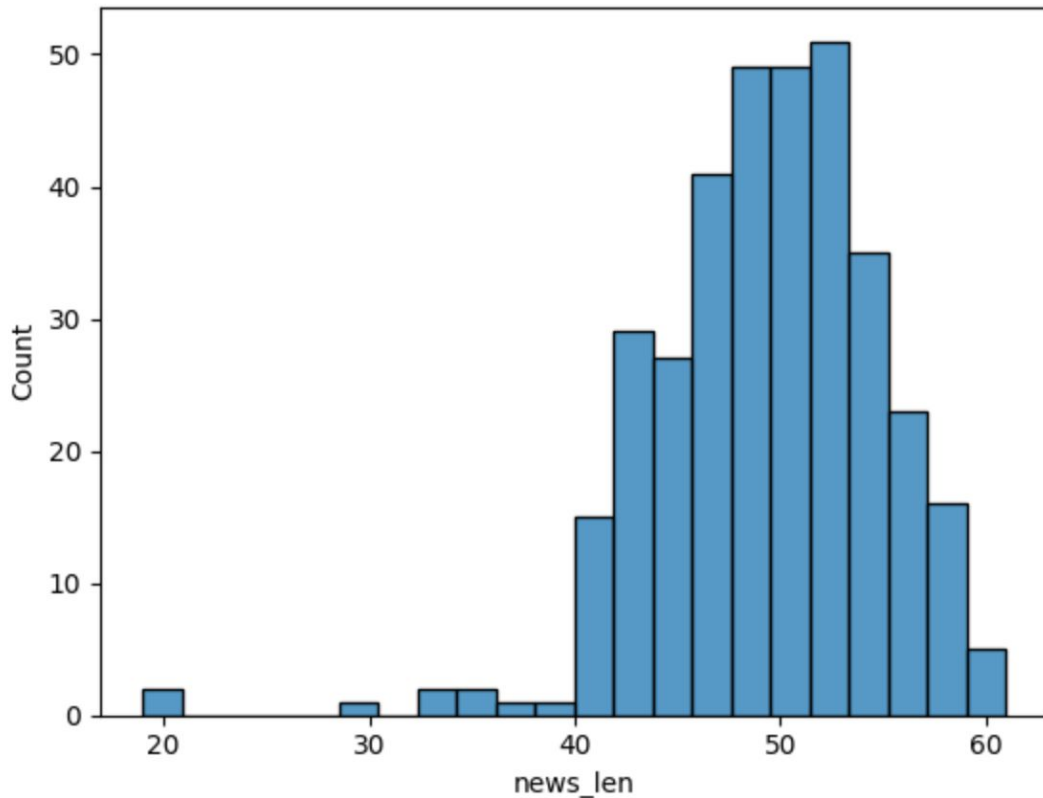
The distributions of are quite similar, indicating a strong correlation between these price metrics. Exhibit binomial distribution. The peaks occurs around the low 40s and the high 50s to low 60s. Right skewed.

EDA: Volume



- Right-skewed distribution, indicating that lower trading volumes are more frequent than higher volumes.
- Volumes concentrated between approximately 80 million and 160 million shares.

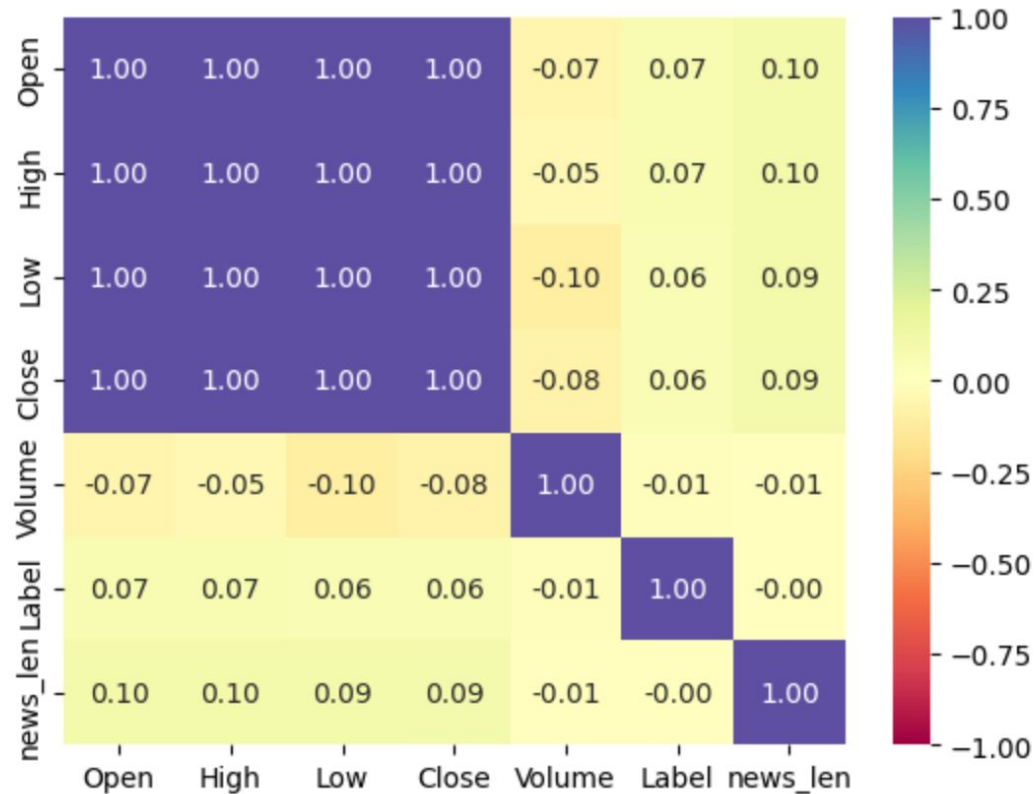
EDA: News length



	news_len
count	349.000000
mean	49.312321
std	5.727770
min	19.000000
25%	46.000000
50%	50.000000
75%	53.000000
max	61.000000

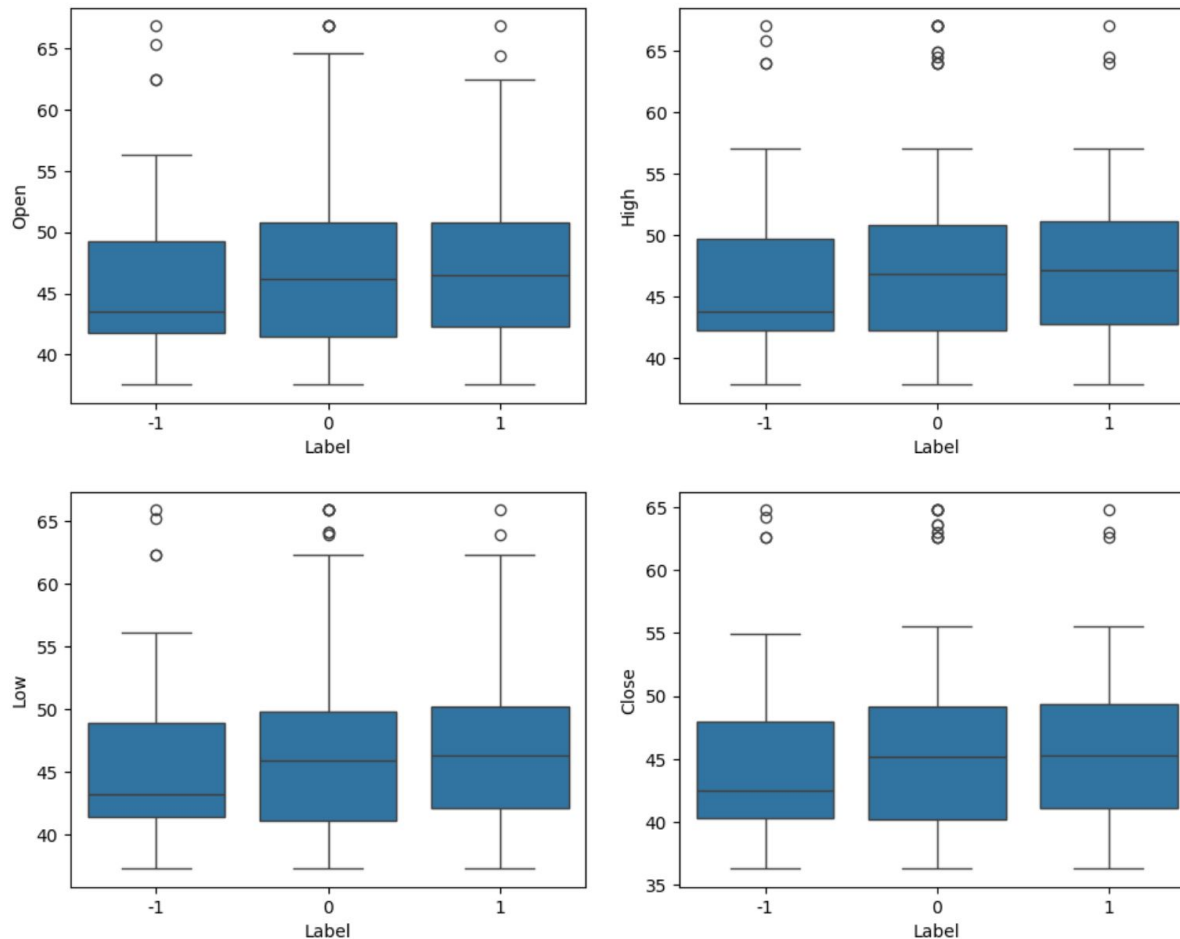
- 49 words on average for a news article

EDA: Correlation matrix



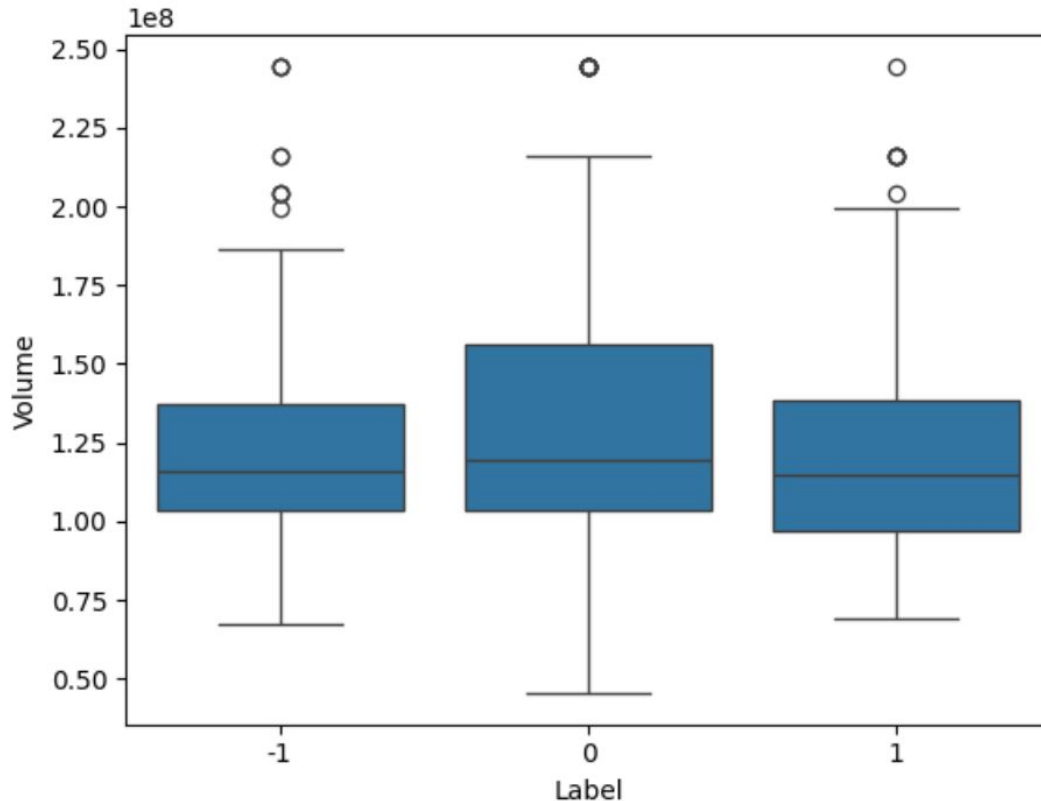
- **High Correlation among Price Metrics**
- **Low Correlation with Volume**
- **Sentiment (Label) and News Length (news_len):** exhibit very low correlation with all other numerical features

EDA: Label Vs Price



There isn't a strong visual indication that a positive news sentiment consistently leads to higher average stock prices, or that negative news consistently leads to lower average stock prices, within the observed daily ranges.

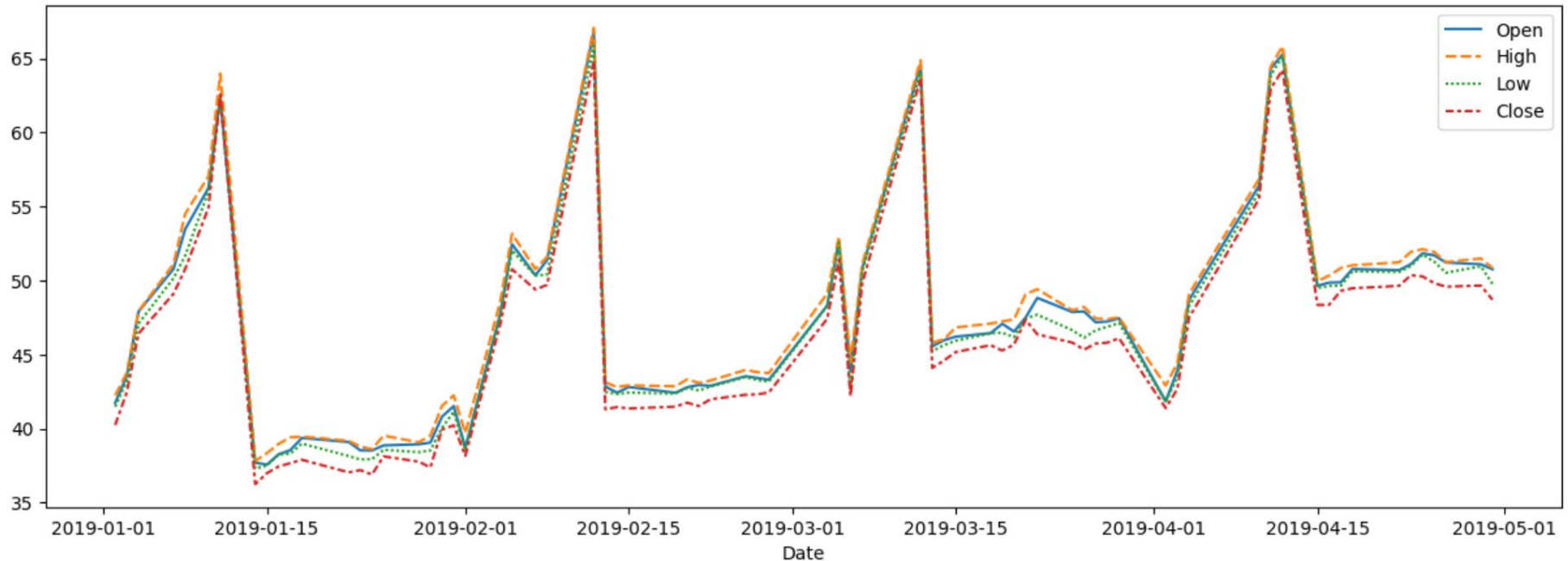
EDA: Label Vs Volume



Weak relationship between Sentiment and Volume

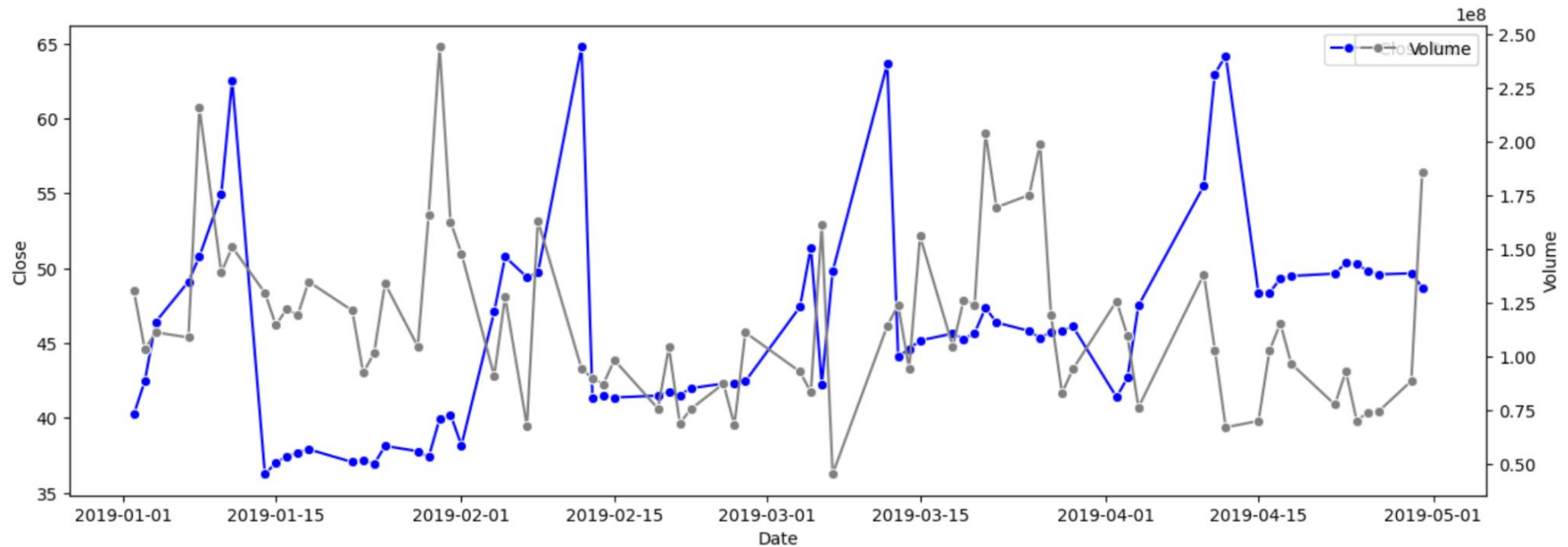
Similar to the price metrics, there doesn't appear to be a strong direct relationship between the categorical news sentiment and the total trading volume.

EDA: Date Vs Price



- The stock generally starts the period (January 2019) at a certain level, dips slightly, then experiences a significant upward trend towards the end of the period (April 2019), reaching higher price points.
- All four price metrics (Open, High, Low, Close) track each other very closely, as expected.

EDA: Date Vs Volume



- Volume (The number of shares traded) is less directly correlated with the Stock price.

Text vectorisation

- **Word embeddings:**
 - Word2Vec: Predict word from context
 - GloVe: Combines local context with global corpus
 - SBERT: Semantics and semantic relevance using Transformers

Text vectorisation

Create dataframes of vectorised content

- Word2Vec

```
▶ print(X_train_wv.shape, X_val_wv.shape, X_test_wv.shape)
```

```
... (286, 300) (21, 300) (42, 300)
```

- Glove

```
print(X_train_gl.shape, X_val_gl.shape, X_test_gl.shape)
```

```
(286, 100) (21, 100) (42, 100)
```

- SBERT

```
print(X_train_st.shape, X_val_st.shape, X_test_st.shape)
```

```
(286, 384) (21, 384) (42, 384)
```

Model Building

- Decision Tree Classifier
- Random Forest Classifier
- Gradient Boosting Classifier

Model Performance

Decision Tree Classifier

	Model	Accuracy	Recall	Precision	F1-Score
1.	Word2Vec	0.23	0.23	0.25	0.24
2.	Glove	0.42	0.42	0.45	0.43
3.	SBERT	0.52	0.52	0.44	0.48

Model Performance

Random Forest Classifier

	Model	Accuracy	Recall	Precision	F1
1.	Word2Vec	0.38	0.38	0.31	0.34
2.	Glove	0.47	0.47	0.4	0.42
3.	SBERT	0.61	0.61	0.53	0.50

Model Performance

Gradient Boosting Classifier

	Model	Accuracy	Recall	Precision	F1
1.	Word2Vec	0.57	0.57	0.43	0.49
2.	Glove	0.38	0.38	0.32	0.35
3.	SBERT	0.47	0.47	0.3	0.36

Model Performance summary

- SBERT embeddings generated high scores for Decision Tree and Random Forest.
- Word2Vec generated high score for Gradient Boosting.
- Random Forest classifier with SBERT produced a higher F1 Score of 0.5 followed by Decision Tree of 0.48

Model Performance

Random Forest Tuned SBERT

	Model	Accuracy	Recall	Precision	F1
1.	Tuned SBERT	0.428571	0.428571	0.542857	0.417989

Conclusion

1. **News Extraction and Categorization:** The model successfully processed weekly news articles and categorized key events into 'Positive Events' and 'Negative Events' as structured lists. This demonstrates the effectiveness of the LLM in understanding and segmenting news sentiment.
2. **Sentiment Trend Visualization:** The weekly sentiment score has been successfully visualized, showing trends, peaks, and troughs in market sentiment derived from news events over time.
3. **Correlation with Stock Price:** A correlation analysis between the 'Sentiment Score' and 'Weekly Average Close Price' yielded a coefficient of -0.16 , indicating a very weak negative linear relationship. This suggests that while there is a slight tendency for stock prices to decrease as sentiment improves, the relationship is not strong.

Part 2: News summarisation

- The model is expected to summarize the news from the week by identifying the top three positive and negative events that are most likely to impact the price of the stock.
- As an output, the model is expected to return a file containing two keys, one for Positive Events and one for Negative Events.

New summarisation

- **Model Loading:**

It downloads and loads the

Mistral-7B-Instruct-v0.2-GGUF model from Hugging Face, configured for GPU usage.

Data pre-processing

- Converting the 'Date' column to datetime objects
- **Data Aggregation:** concatenating all news for each week into a single string.

	Date	News
0	2019-01-06	The tech sector experienced a significant dec...
1	2019-01-13	Sprint and Samsung plan to release 5G smartph...
2	2019-01-20	The U.S. stock market declined on Monday as c...
3	2019-01-27	The Swiss National Bank (SNB) governor, Andre...
4	2019-02-03	Caterpillar Inc reported lower-than-expected ...
5	2019-02-10	The Dow Jones Industrial Average, S&P 500, an...
6	2019-02-17	This week, the European Union's second highes...
7	2019-02-24	This news article discusses progress towards ...
8	2019-03-03	The Dow Jones Industrial Average and other ma...

Prompt Engineering

1. **Role:** You are an expert data analyst specializing in news content analysis.
2. **Task:** Analyze the provided news headline and return the main topics contained within it.
3. **Instructions:**
 01. Read the news headline carefully.
 02. Identify the main subjects or entities mentioned in the headline.
 03. Determine the key events or actions described in the headline.
 04. Extract relevant keywords that represent the topics.
 05. List the topics in a concise manner.
4. **Return the output in JSON format** with keys as the topic number and values as the actual topic. **Sample Output:** {"1": "Politics", "2": "Economy", "3": "Health" }

Response function

1.	max_tokens: maximum number of tokens (which can be words or sub-word units) that the (LLM) is allowed to generate in its response. If the generated output exceeds this limit, it will be truncated.	4,000
2.	temperature: This parameter controls the randomness and creativity of the LLM's output. It influences the probability distribution of the next token selection.	0.7 A very low temperature might lead to overly generic or repetitive summaries, while a very high one could risk 'hallucinating' or generating irrelevant information.
3.	top_p: This parameter controls the diversity of the generated text by considering a subset of tokens whose cumulative probability exceeds a certain threshold p.	0.9 A top_p of 0.9 is a common and effective choice for striking a good balance between diversity and quality.
4.	top_k (Top-K Sampling): This parameter limits the number of highest probability tokens that the model considers for the next word prediction.	top_k=50 helps in controlling the fluency and coherence of the generated text. It ensures that the model considers a reasonable number of high-probability tokens at each step of generation, preventing it from getting stuck on less suitable words.

Response function

```
▶ %%time
summary = response_mistral_1(prompt, news)
print(summary)

... {
    "Positive Events": [
        "Roku Inc announced plans to offer premium video channels on a subscription basis through its free streaming service, The Roku Channel.",
        "The Supreme Court will review Broadcom's appeal in a shareholder lawsuit over the 2015 acquisition of Emulex.",
        "The Chinese central bank announced a fifth reduction in the required reserve ratio (RRR) for banks, freeing up approximately 116.5 billion yuan for new
    ],
    "Negative Events": [
        "Apple lowered its fiscal Q1 revenue guidance to $84 billion from earlier estimates of $89-$93 billion due to weaker than expected iPhone sales.",
        "Apple cut its fiscal first quarter revenue forecast from $89-$93 billion to $84 billion due to weaker demand in China and fewer iPhone upgrades.",
        "Apple's revenue warning led to a decline in USD JPY pair and a gain in Japanese yen, as investors sought safety in the highly liquid currency."
    ]
}
CPU times: user 27 s, sys: 6.98 s, total: 34 s
Wall time: 34.3 s
```

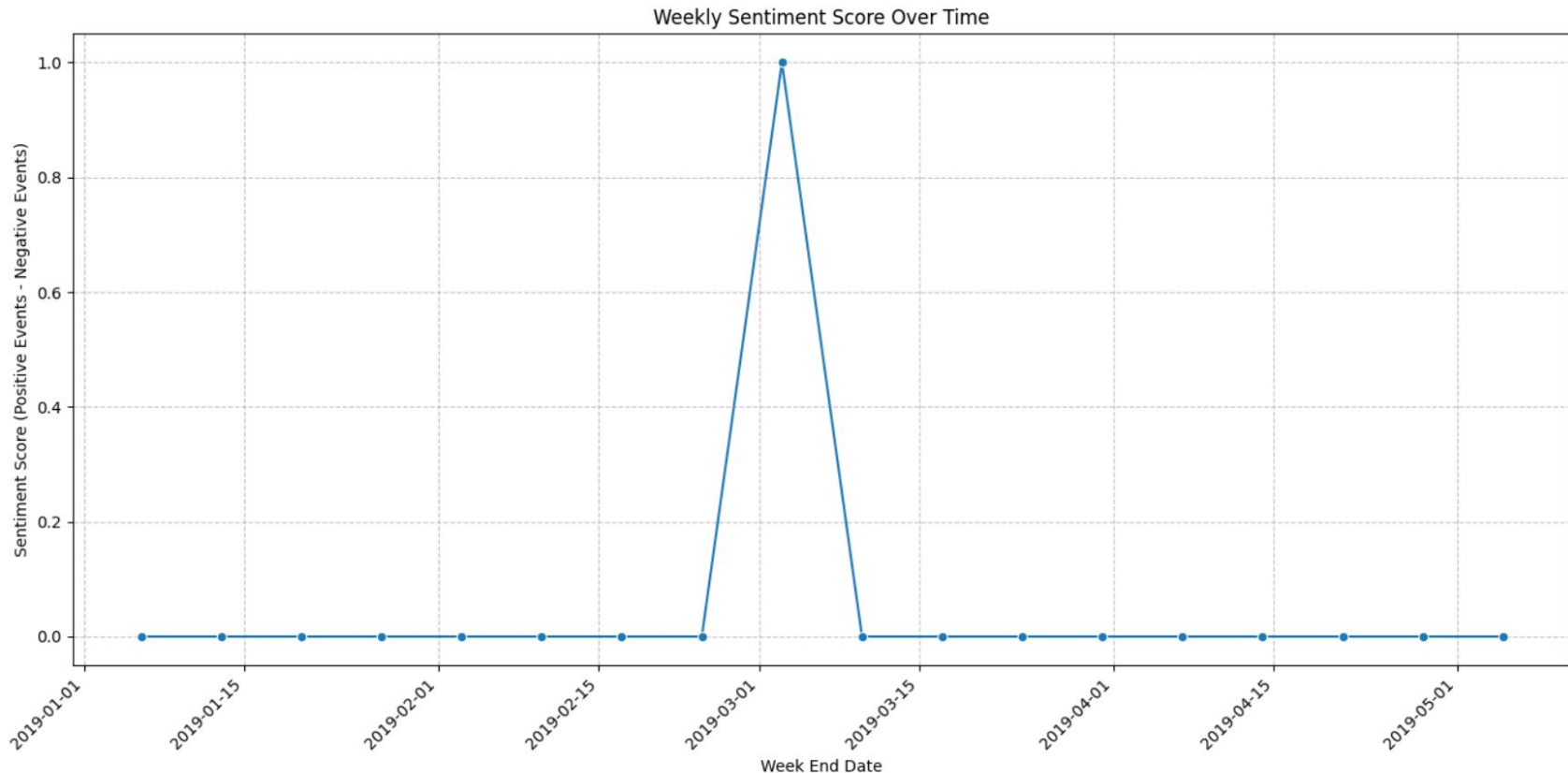
News Article

	Week End Date	News	Week Positive Events	Week Negative Events
0	2019-01-06	The tech sector experienced a significant dec...	[Roku Inc announced plans to offer premium vid...	[Apple lowered its fiscal Q1 revenue guidance ...
1	2019-01-13	Sprint and Samsung plan to release 5G smartph...	[Sprint and Samsung plan to release 5G smartph...	[Geely forecasts flat sales for 2019 due to ec...
2	2019-01-20	The U.S. stock market declined on Monday as c...	[Apple CEO Tim Cook announced internal staffin...	[Concerns over global economic slowdown intens...
3	2019-01-27	The Swiss National Bank (SNB) governor, Andre...	[SNB governor Andrea Maechler emphasized the i...	[White House economic adviser Lawrence Kudlow ...
4	2019-02-03	Caterpillar Inc reported lower-than-expected ...	[Apple reported stronger-than-expected earning...	[Caterpillar Inc reported lower-than-expected ...

News Article

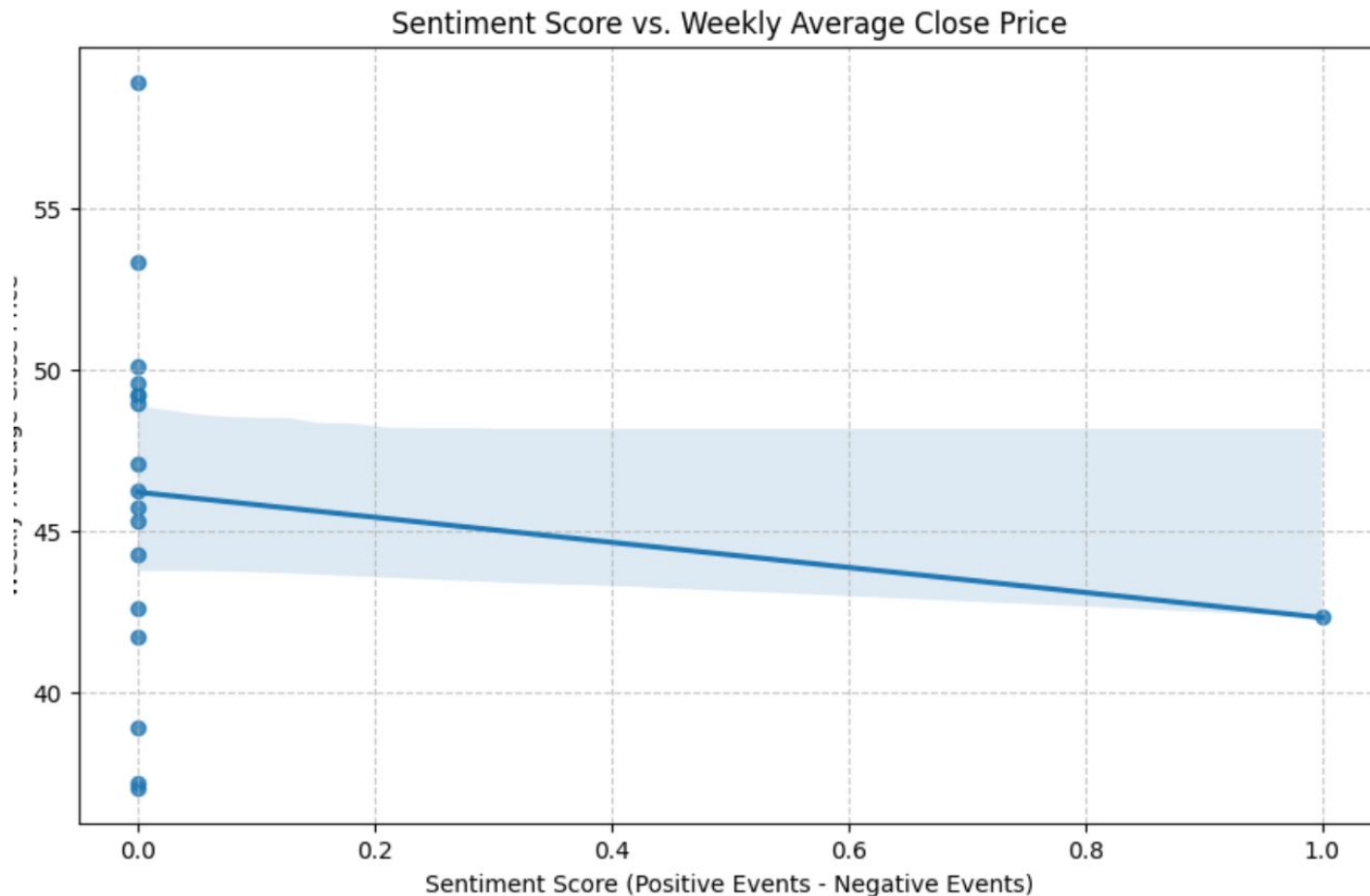
	Week End Date	News	Week Positive Events	Week Negative Events	Sentiment Score
0	2019-01-06	The tech sector experienced a significant dec...	[Roku Inc announced plans to offer premium vid...	[Apple lowered its fiscal Q1 revenue guidance ...	0
1	2019-01-13	Sprint and Samsung plan to release 5G smartph...	[Sprint and Samsung plan to release 5G smartph...	[Geely forecasts flat sales in 2019 due to eco...	0
2	2019-01-20	The U.S. stock market declined on Monday as c...	[Apple CEO Tim Cook announced internal staffin...	[The U.S. stock market declined on Monday due ...	0
3	2019-01-27	The Swiss National Bank (SNB) governor, Andre...	[Swiss National Bank governor Andrea Maechler ...	[The Dow, S&P 500, and Nasdaq experienced sign...	0
4	2019-02-03	Caterpillar Inc reported lower-than-expected ...	[Apple reported higher-than-expected Q4 earnin...	[Caterpillar reported lower-than-expected Q4 e...	0

News Article



- A positive score indicates that more positive events were identified than negative events, suggesting a generally optimistic sentiment.
- A negative score suggests more negative events than positive ones, indicating a pessimistic sentiment.
- A score of zero (as observed in some initial weeks) means an equal number of positive and negative events were found, or no significant events were identified.

News Article



The correlation coefficient is **-0.16**, indicating a very weak negative linear relationship between the two variables. This suggests that as the sentiment score increases, the weekly average close price tends to slightly decrease, but the relationship is not strong.

Conclusion

1. **News Extraction and Categorization:** The model successfully processed weekly news articles and categorized key events into 'Positive Events' and 'Negative Events' as structured lists. This demonstrates the effectiveness of the LLM in understanding and segmenting news sentiment.
2. **Sentiment Trend Visualization:** The weekly sentiment score has been successfully visualized, showing trends, peaks, and troughs in market sentiment derived from news events over time.
3. **Correlation with Stock Price:** A correlation analysis between the 'Sentiment Score' and 'Weekly Average Close Price' yielded a coefficient of -0.16 , indicating a very weak negative linear relationship. This suggests that while there is a slight tendency for stock prices to decrease as sentiment improves, the relationship is not strong.
4. **Initial Sentiment Score:** For the initial weeks shown, the 'Sentiment Score' is zero. This indicates that the number of identified positive events is equal to the number of negative events for those specific weeks. This could mean a balanced news week, or that the model found an equal number of influential events on both sides.

Recommendations

1. **Qualitative analysis of events:** While we have lists of positive and negative events, a deeper dive into the content of these events (beyond just their count) would provide richer insights. For instance, understanding why certain events are deemed positive or negative and their potential magnitude of impact.
2. **Refine event significance:** Currently, each event contributes equally to the sentiment score. Future work could involve assigning weights or impact scores to events based on their perceived significance to provide a more nuanced sentiment analysis.
3. **Explore other correlations:** Investigate correlations with other financial indicators or market segments.

Skills Demonstrated

- Machine Learning & Model Evaluation
- Natural Language Processing
- Text Analysis
- Python & scikit-learn
- LLMs
- Sentiment Analysis

Thank You

- Thank you for your time
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